Digital Logic Circuits (Spring 2026)

HW #1

19. What are two ways of representing zero in 1's complement form?

20. How is zero represented in 2's complement form?

19. Zero is represented in 1's complement as all 0's (for +0) or all 1's (for −0).

20. Zero is represented by all 0's only in 2's complement.

24. Express each decimal number as an 8-bit number in the 1's complement form:

(a) −34 **(b)** +57 **(c)** −99 **(d)** +115

25. Express each decimal number as an 8-bit number in the 2's complement form:

(a) +12 **(b)** −68 **(c)** +101 **(d)** −125

24. (a) Magnitude of 34 = 0100010
−34 = 11011101

(b) Magnitude of 57 = 0111001
+57 = 00111001

(c) Magnitude of 99 = 1100011
−99 = 10011100

(d) Magnitude of 115 = 1110011
+115 = 01110011

25. (a) Magnitude of 12 = 1100
+12 = 00001100

(b) Magnitude of 68 = 1000100
−68 = 10111100

(c) Magnitude of 101_{10} = 1100101
$+101_{10}$ = 01100101

(d) Magnitude of 125 = 1111101
−125 = 10000011

29. Express each of the following sign-magnitude binary numbers in single-precision floating-point format:

(a) 0111110000101011 **(b)** 100110000011000

30. Determine the values of the following single-precision floating-point numbers:

(a) 1 10000001 01001001110001000000000

(b) 0 11001100 10000111110100100000000

29. (a) 0111110000101011 $\rightarrow$ sign = 0

$1.11110000101011 \times 2^{14} \rightarrow$ exponent = 127 + 14 $\boxed{+}$ 141 = 10001101

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Mantissa = 11110000101011000000000

01000110111110000101011000000000

(b) 100110000011000 $\rightarrow$ sign = 1

$1.10000011000 \times 2^{11} \rightarrow$ exponent = 127 + 11 = 138 = 10001010

Mantissa = 10000011000000000000000

11000101011000001100000000000000

30. (a) 11000000101001001110001000000000

Sign = 1

Exponent = 10000001 = 129 − 127 = 2

Mantissa = $1.01001001110001 \times 2^2$ = 101.001001110001

−101.001001110001 = **−5.15258789**

(b) 01100110010000111110100100000000

Sign = 0

Exponent = 11001100 = 204 − 127 = 77

Mantissa = 1.100001111101001

$\mathbf{1.100001111101001 \times 2^{77}}$

33. Perform each addition in the 2's complement form:

(a) 10001100 + 00111001 **(b)** 11011001 + 11100111

34. Perform each subtraction in the 2's complement form:

(a) 00110011 − 00010000 **(b)** 01100101 − 11101000

35. Multiply 01101010 by 11110001 in the 2's complement form.

33. (a)

$$\begin{array}{r} 10001100 \\ +\,00111001 \\ \hline 11000101 \end{array}$$

(b)

$$\begin{array}{r} 11011001 \\ +\,11100111 \\ \hline 11000000 \end{array}$$

34. (a)

$$\begin{array}{r} 00110011 \\ -\,00010000 \\ \hline \end{array} \qquad \begin{array}{r} 00110011 \\ +\,11110000 \\ \hline \cancel{1}\;00100011 \end{array}$$

(b)

$$\begin{array}{r} 01100101 \\ -\,11101000 \\ \hline \end{array} \qquad \begin{array}{r} 01100101 \\ +\,00011000 \\ \hline 01111101 \end{array}$$

35.

$$\begin{array}{r} 01101010 \\ \times\,11110001 \\ \hline \end{array} \qquad \begin{array}{r} 01101010 \\ \times\,00001111 \\ \hline 01101010 \\ 01101010 \\ \hline 100111110 \\ 01101010 \\ \hline 1011100110 \\ 01101010 \\ \hline 11000110110 \end{array}$$

Changing to 2’s complement with sign: 100111001010